

- Project Summary: Ren Yuan
 - Who we are
 - Functional Description (Concise)
 - Overview
 - Receive command from PC
 - Identify ANNN/L/P pattern
 - Processing, echoing
 - Return the result
 - Data Processor
 - Find the peak value among generated data.
 - Command Processor
 - ANNN detect
 - Control the Rx. Tx. & DP.
- Project Charter: Ren Yuan
 - In Scope
 - DP, CP development
 - From algorithm to HDL,
 - From simulation to synthesis & board testing
 - System integration with provided modules
 - Documentation
 - Out of Scope
 - Data Generator
 - Testbenches
 - Mainly from university
 - Test some sub-modules, not formally, no requirement for coverage.
 - PPA
 - Power, Area: ignore.
 - Performance: only consider latency
 - Constraints
 - 100MHz frequency at minimum
 - Specified board
 - Compatibility with provided modules
 - Deadlines
 - Deliverables
 - FPGA Peak Detector source code
 - Group Report
 - Video on group collaboration
 - Target
 - Learn digital design process, industrial tools...
 - Learn **group collaboration**

Dictate of p1, p2: Detailed organisation is upcoming sections.....

- Team Structure: Xi Zhao
 - Yuan Ren (Data Processor Team)
 - Lead
 - VHDL implementation
 - Code review
 - Report documentation
 - Support
 - Simulation and debugging
 - Julia Yang (Data Processor Team)
 - Lead
 - FSM/ASM design
 - Simulation and debugging
 - Report Writing (DP part)
 - Support
 - VHDL code review
 - System integration on board
 - Xinrui Zhu (Command Processor Team)
 - Lead
 - System architecture design
 - Core FSM/ASM design
 - System integration on board
 - Meeting organisation
 - Support
 - VHDL implementation
 - Code review
 - Yuhao Jiang (Command Processor Team)
 - Lead
 - IO controller design
 - VHDL implementation
 - Support
 - Simulation and debugging
 - Report documentation (CP part)
 - Xi Zhao (Command Processor Team)
 - Lead
 - Simulation and debugging
 - Report Writing (CP part)
 - Progress tracking

- Support
 - Code review

Dictate of p3:

DP team includes Yuan Ren, Julia. meanwhile, CP team ...

Why is the responsibility divided in such a way?

- Planning:
 - Work Breakdown Structure
 - Stage1: Algorithm, FSM, ASM charts
 - Stage2: Implementation in VHDL
 - Stage3: System integration and cross-testing
 - Stage4: Vivado synthesis, implementation, board testing
 - Stage5: Video, report
 - Timeline
 - Milestones
 - Timeline
 - Risk control
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Dictate: How these tools are useful for task planning.

Limitation:

Unfamiliar with task requirement in TB1.

Unfamiliar with FPGA prototyping procedure in TB1.

...

开会时间没说。。。。

- Executing: XR
 - Coordination
 - Meetings in person
 - Monday Session: CP Group
 - Tuesday Session: DP Group
 - Teams: Online meeting
 - Tuesday night: CP
 - Wednesday night: DP
 - GitHub: version control
 - They write, I push.
 - Everyone reviews
 - Progress tracking
 - Orally Report on sessions
 - Communicate on Instant Messaging Software
 - On GitHub
 - Code blame
 - Pull requests: Commit and File Changes
 - Log by commitments: what's accomplished and what's next
 - Issues and Responses
 - Technical
 - Forget VHDL, no experience using tools like Vivado
 - Limited GitHub experience
 - Misunderstand task requirement
 - Difference between plan and actual
 - Algorithm & Simulation: Longer than expected
 - Assemble: Shorter
 - Video & Report: details ignored in plan
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 - Member availability
 - During Easter...
 - Incidents

- Outcomes: Julia

- Product

- Satisfy the requirement
 - Enhanced echo function: ANNN Optimization (Will not be interrupted when input)
 - Higher Frequency than expected 100MHz
 - CP:
 - DP:

- Team collaboration

- Regular meeting
 - Equally contributed

- Lessons learned: Julia

- Summary

- Scheduling approaches
 - Gantt Chart
 - Work Distribution
 - Risk register
 - Collaboration approaches
 - Tools: GitHub, Teams, Mail CC/BCC
 - Progress tracking
 - Technical
 - Digital design methodology: VHDL syntax, FSM/ASM
 - Tool chains: Intel Questa, Xilinx Vivado Design Suite

- The Future

- Better interpretation of task requirement
 - More detailed group plan
 - Because video and report are missing
 - ...
 - Communication efficiency
 - Take actions after meeting